25900975 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR /IMPROVEMENT/ REAPPEARANCE EXAMINATION NOVEMBER 2025

FIRST SEMESTER

Discipline Specific Core Course (DSC) - MG1DSCSAC100 - INTRODUCTION TO STRENGTH AND CONDITIONING

(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer **All** Questions

Each Question carries **1** mark

1. Which skill related component of fitness is most important for a sprinter? [U] / [CO2]

   a). Power b). Balance

   c). Flexibility d). Muscular Endurance

2. What equipment is often used in functional training? [K] / [CO4]

   a). Only body weight b). Kettlebells, medicine balls, and elastic bands

   c). Heavy weight machines exclusively d). None; no equipment is needed

3. Which of the following is considered a functional flexibility exercise? [U] / [CO1, CO2]

   a). Holding a hamstring stretch for 30 seconds b). Performing leg swings and arm circles

   c). Sitting in a butterfly stretch d). Doing calf raises

4. What does tactical training primarily involve? [U] / [CO3, CO4]

a). Developing physical endurance
b). Understanding and applying strategies during competition
c). Enhancing speed and agility
d). Learning the rules of the game

5. Who should give the final clearance for an athlete to return to play? [U] / [CO2, CO4]

a). The Athlete's Coach
b). The Athlete
c). A Medical Professional
d). The Athlete's Teammates

6. Body Mass Index (BMI) primarily measures: [U] / [CO2, CO5]

a). Body composition
b). Height and weight
c). Muscle density
d). Vo2 max

7. What is a resting heart rate below 60 bpm commonly referred to as? [K] / [CO2]

a). Tachycardia
b). Bardycardia
c). Hypertension
d). Hypotension

8. During the rehabilitation process, eccentric exercises are used primarily to: [An] / [CO2]

a). Strengthen muscles without increasing flexibility
b). Focus on the concentric phase of the muscle contraction
c). Improve muscle strength while protecting healing tissues
d). Increase aerobic endurance

9. What is the primary goal of biomechanics in sports and physical exercise? [U] / [CO3, CO4]

a). To increase muscle mass
b). To improve performance and increase physical fitness
c). To enhance flexibility
d). To reduce body weight

10. In skill integration, which method is commonly used to teach athletes complex movements? [U] / [CO2]

a). Isolated strength training
b). Task-oriented practice
c). Passive stretching
d). Static conditioning

[10x1 = 10]

Part B

Short Answer Type Questions

Answer any **10** Questions

Each Question carries **2** marks

11.	Name one of the primary components of modern strength and conditioning programs.	[K] / [CO4]
12.	Analyze the importance of communication and collaboration between strength and conditioning coaches and other members of the sports performance team (e.g., head coach, physiotherapist, nutritionist).	[An] / [CO2, CO4]
13.	What role does injury prevention play in the Long-Term Athlete Development (LTAD) framework?	[U] / [CO1]
14.	What is the role of a strength and conditioning coach?	[U] / [CO1]
15.	Can strength and conditioning programs be beneficial for older adults regarding cardiovascular health?	[U] / [CO1, CO2, CO4]
16.	What might happen if an athlete does not get sufficient recovery after the alarm stage of General Adaptation Syndrome (GAS)?	[An] / [CO2]
17.	How can body composition be assessed?	[U] / [CO2, CO4]
18.	What is the difference between flexibility and mobility, and why is each important for athletes?	[U] / [CO1]
19.	How would you design a training program using the SAID principle for a long-distance runner?	[C] / [CO2, CO3, CO4]
20.	What is conditioning, and how does it relate to athletic performance?	[U] / [CO1]
21.	What significant trend in strength and conditioning emerged in the late 20th century related to sports science?	[An] / [CO2]
22.	What is muscular imbalance, and how can it contribute to injury?	[U] / [CO1, CO4]

[10x2 = 20]

Part C

Short Essay Type Questions

Answer any **4** Questions

Each Question carries **5** marks

23.	Define the components of skill-related physical fitness and explain their relevance to athletic performance.	[An] / [CO1]
24.	What are the methods for assessing body composition?	[U] / [CO4]

25. Define the components of health-related physical fitness and explain their importance for overall well-being. [U] / [CO1]

26. Discuss the considerations for strength and conditioning programs for special populations, such as older adults, individuals with disabilities, or those with chronic health conditions. [U] / [CO2, CO3, CO4]

27. Explain the SAID Principle and its significance in exercise science. [U] / [CO4]

28. How does a proper warm-up routine contribute to injury prevention? [U] / [CO1, CO4]

[4x5 = 20]